Module details of the implemented model

Residual paths, normalization order and output constraint

A Equivariant graph block × 6

$v_{out} = v' \odot \mathrm{sigmoid}(W_g\, h_{out})$

$h_{out} = \mathrm{LayerNorm}(h' + \mathrm{FFN})$

FFN([h′; ‖v′‖])
304 → 512 → 256
SiLU + Dropout

Residual message updates
$h' = h + \mathrm{Dropout}(W_s\, \Sigma m_s)$
$v' = v + W_v\, \Sigma m_v$

Edge-conditioned attention
softmax over incoming neighbors
Scalar + directional vector messages

LayerNorm(h) → Q, K, V

Scalar h: N × 256; vector v: N × 48 × 3

B Pre-LN Transformer block

Descriptor × 4 | Adapter × 2
Fusion × 4; final LayerNorm after each stack

Residual add (+ dropout)

Feed-forward network
256 → 1024 → 256 · GELU

LayerNorm

Residual add (+ dropout)

Multi-head self-attention
8 heads · d = 256

LayerNorm

Input tokens X

C Gas-wise output transform

No Softmax on the regression output

5 nonnegative uptake values / gas
$q(p_1) \le q(p_2) \le \cdots \le q(p_5)$

expm1: q = exp(log(1 + q)) − 1

Upper clamp at 6

Cumulative sum
Increasing log(1 + q)

Softplus
Positive log-space increments

10 raw MLP outputs
Split into CO_2 (5) and CH_4 (5)

Attention uses Softmax internally. The final adsorption output uses a cumulative Softplus transform, not a probability normalization.